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MILITARY SPECIFICATION

MANUALS, TECHNICAL, AND CHECKLIST: NUCLEAR DELIVERY, LEVEL, STRATEGIC BOMBER AIRCRAFT, PREPARATION OF

1. SCOPE

1.1 This specification covers the requirements for the preparation of technical manuals and checklists for the level delivery of nuclear weapons from strategic bomber aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-38778	Checklists: Title page, List of Effective Pages, Printing and Binders; General Requirements for Preparation Of
MIL-M-38784	Manuals, Technical: General Requirements for Preparation Of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for (Photolithographic Negatives, Reproduction Assembly, Printing)

PUBLICATIONS

Technical Orders

00-5-1	AF Technical Order System
00-5-2	Technical Order Distribution and Requisition System

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements

3.1.1 General specifications. The requirements of MIL-P-38790, MIL-C-38778, and MIL-M-38784 apply as requirements of this specification to the extent specified herein. Unless otherwise specified in the contract, the technical manual and checklist shall be submitted in the form of negatives.

3.1.1.1 Lists. Lists of tables, illustrations, and TCTOs shall not be required.

3.1.1.2 Illustrations. Illustrations may be used more than once, provided such use is essential. Reference in the text to illustrations need not state specifically the purpose of the illustration if the purpose is obvious. Figure titles may be placed within illustrations; the figure number (and classification markings, if applicable) shall then be placed below the illustration. Figure numbers and titles shall not be used in the normal aircrew procedures section. If an illustration is required for a specific step in the procedures, the step number shall be placed below the lower left corner of the illustration.

3.1.1.3 Revisions. Change bars shall not be required in revisions.

3.1.1.4 Bomb tables. Bomb tables shall not carry table numbers; reference to bomb tables shall be to the applicable page numbers. If ballistic graphs are used instead of bomb tables, the graphs shall carry figure numbers. Black, bleed tabs shall be used to identify bomb tables (ballistic graphs); the tables (graphs) shall then be preceded by an index page identifying the tabs. Tabs shall be no smaller than 1/2-inch high by 1/4-inch wide.

3.1.1.5 Identification of procedural and bomb table (ballistic graph) pages.

a. Bomb (or Bomb Pod) Preflight Procedures. These pages shall carry the bomb designation and delivery mode immediately below the TO number.

b. Bomb Tables (Ballistic Graphs). These pages shall carry the bomb designation and delivery mode immediately below the TO number.

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3.1.1.6 Nomenclature. Nomenclature shall be standardized throughout the manual. Nomenclature may be shortened or abbreviated to its common name. The following policy shall apply to items other than nuclear weapon (bomb) components and functions. When referring to aircraft components or functions, the normal dash 1 (flight manual) nomenclature shall be used.

3.1.1.7 Schematics. Schematics and symbols on schematics need not conform to the standard cited in MIL-M-38784. Color may be used for schematic clarification.

3.1.1.8 Text Presentation. Unless otherwise specified in the contract text shall be presented single column, with justification of the right margin optional. Standard office typewriter may be used to prepare reproducible copy. Final type size within a manual shall be approximately 10 point (pt.) Artwork shall be integrated with the text.

3.1.1.9 Flagging. Flags shall be used in the TOs (manuals and checklists), if required, to indicate daily checks, or to denote aircraft models and bomb/pod types.

3.1.1.10 Abbreviations. Only nonstandard abbreviations shall be explained in the introduction to the manual.

3.2 Arrangement. The technical order shall have the following arrangement:

Volume I - Basic Information
Volume II - Bomb (or weapon, as applicable) Operating Procedures
Volume III - Ballistics
Checklist(s)

Manuals shall be standard size (8-1/4 by 10-3/4 inches); checklist(s) shall be 4-1/2 by 7 inches, trim size.

3.2.1 Front matter (Volumes I, II, and III). The front matter for each volume of the manual shall consist of the following items:

Title Page
List of Effective Pages
Table of Contents
Introduction

3.2.1.1 Title page. The title page for each manual shall be prepared in accordance with the requirements of MIL-M-38784. One of the following transmittal notices (distribution limitation statements) shall be placed on the title page as indicated in MIL-M-38784.

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a. For Classified TOs: "In addition to security requirements which apply to this document and which must be met, each transmittal outside the Department of Defense must have prior approval of the Technical Order Distribution Control Activity. Refer to TO 00-5-2."

b. For Unclassified TOs: "Each transmittal of this document outside the Department of Defense must have prior approval of the Technical Order Distribution Control Activity. Refer to TO 00-5-2."

3.2.1.2 List of effective pages. The list of effective pages shall be prepared in accordance with the requirements of MIL-M-38784.

3.2.1.3 Table of contents. Each volume shall contain a table of contents prepared in accordance with the general requirements of MIL-M-38784. In addition, each section shall have a table of contents listing the major headings within the section. When the table of contents is brief (less than one half of a page), the introduction or section text shall begin immediately below the table of contents. Tables of contents shall be typed single column.

3.2.1.4 Introduction. The introduction shall be prepared in accordance with the requirements of MIL-M-38784. Also the following additional requirements apply:

a. List of applicable reference documents. This list shall be held to a minimum, showing essential documents only, and may be included in only one volume of a multivolume series.

b. Statements concerning technical responsibility for the manual and submission of reports and comments.

3.2.2 Detail requirements for volume I (basic information). Volume I shall consist of the following in the order indicated:

Front Matter
Section I - Bomb Monitor and Control System
Section II - Bomb Suspension and Release Systems
Section III - Bomb Description
Section IV - Nuclear Bomb Effects

3.2.2.1 Front matter. Front matter shall be prepared as specified in 3.2.1 herein.

3.2.2.2 Section I - Bomb monitor and control system. This section shall contain information to locate, identify, and functionally describe the components of the bomb monitor and control system of direct concern to the aircrews. Schematics may be included.

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3.2.2.3 Section II - Bomb suspension and release system. This section shall contain information to locate, identify, and functionally describe the bomb suspension and release systems of direct concern to the aircrew. Both normal and emergency release systems, with their associated equipment and panels, and all suspension systems, shall be covered. Schematics may be included.

3.2.2.4 Section III - Bomb description. This section shall contain bomb descriptions and schematics. Data shall be presented separately for each bomb except when the descriptions can be combined. Data shall be presented in the following order for each bomb:

a. Bomb Description - Information to locate, identify and functionally describe the bomb and components of direct interest to the aircrews.

b. Simplified drop sequence diagram.

c. Simplified monitoring and control schematic.

d. Simplified fuzing and firing schematic.

3.2.2.5 Section IV - Nuclear bomb effects. This section shall contain nuclear bomb/warhead effects. An explanation of the use of escape tables (or charts) shall be included, along with an example and sample problem. Charts depicting shock arrival times for various bombs, with reference to current escape maneuvers, shall be included as applicable.

3.2.2.6 Alphabetical index. The alphabetical index shall follow the requirements of MIL-M-38784. Content shall be limited to subjects, topics, and paragraph titles; illustration and table titles need not be indexed. Index pages shall normally be typed double column, unless the index is less than one column in length.

3.2.3 Detail requirements for volume II (Bomb, or Weapon, Operating procedures). Volume II shall consist of the following in the order indicated:

Front Matter
Section I - Normal Aircrew Procedures
Section II - Emergency Procedures
Section III - Supplementary Data

3.2.3.1 Front matter. Front matter shall be prepared as specified in 3.2.1 herein. In addition, the introduction shall contain cross references to the other volumes and to the checklists and shall contain definitions of Caution, Warning, and Note.

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3.2.3.2 Section I - Normal aircrew procedures. This section, prepared in the form of illustrated checklists, shall contain procedures for the performance of all functions from the time the aircrew reports to a loaded aircraft through postlanding. The procedures shall cover strike, aborted, and ferry missions. If applicable, recovery procedures shall be included. Each step shall consist of an unclassified demand-response line, followed by an amplification, if required. Amplifications may contain classified information. Only necessary illustrations shall be used; steps shall not be illustrated unless required.

3.2.3.2.1 Amplified checklists shall be included for, but not limited to, the following categories.

Preflight Procedures
  Before Exterior Inspection
  (Bomb) Preflight
  Interior Inspection
Pretakeoff Procedures
Inflight Procedures
  Periodic Checks
  Weapon Preparation for Release (WPR)
  Abort Procedures (after WPR)
  Pre-IP and IP (if applicable)
  Bomb Run
  Post Bomb Release
Descent and Landing Procedures
Ground Safing
Recovery (if required)
Emergency War Order (EWO) Restrike
Ferry Procedures

Headings listed above may be combined (such as Recovery and Ferry Procedures) or augmented, based on system requirements and alinement with applicable flight manual procedures. Each major heading shall be centered; horizontal lines shall be used for separating blocks of procedures--two lines to separate major headings from the preceding steps and one line to separate subheadings within a major heading.

3.2.3.2.2 In Preflight procedures, the (bomb) preflight checks shall be presented for each bomb (or bomb pod) carried on the aircraft. Pages containing the (bomb) preflight checks shall be identified as specified in 3.1.1.5a herein.

3.2.3.2.3 Ferry procedures shall be presented in the same manner and format as strike procedures; amplification for steps identical to strike mission steps shall not be repeated in the ferry procedures.

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3.2.3.2.4 Illustrations in this section shall be as small as practical; every attempt shall be made to make art no wider than 20 picas. Needless repetition shall be avoided. Each step (demand-response line item) shall be placed on one line, if possible. Layout of steps and illustrations shall avoid blank spaces. Step designation numbers shall key illustrations to the applicable steps.

3.2.3.2.5 Flags may be used in this section if necessary. The D-flag shall be used to indicate daily checks; other lettered flags may be used to denote aircraft models or bomb/pod types.

3.2.3.3 Section II - Emergency procedures. This section shall include procedures and policies for emergencies involving the bomb(s) or the aircraft with bomb(s) aboard. The procedures within the section shall contain aircrew corrective action in the event an emergency or malfunction occurs.

3.2.3.3.1 Bomb monitoring and control system malfunction analysis, bomb manual safing procedures, and bomb jettison procedures shall be presented separately for each bomb, or combined when practical. Bomb limitations shall be included. Additional emergency data may be included if required.

3.2.3.3.2 A brief explanation of the use of escape charts (or tables) shall be included. Charts (or tables) and related text on the limitations of the aircraft to nuclear radiation, thermal radiation, and shock waves. The charts (or tables) shall show the relative position of the aircraft to the nuclear blast effects during the escape maneuver.

3.2.3.3.3 Emergency pages shall be marked in accordance with the requirements of MIL-M-38784.

3.2.3.4 Section III - Supplementary data. This section shall include information and procedures relative to inflight safety, normal and inflight emergency notification, firefighting and evacuation, and unscheduled landings. Additional data, not specifically covered in sections I and II, may be included.

3.2.4 Detail requirements for Volume III (ballistics). This volume shall contain ballistic computation procedures, illustrated with a sample problem, to be used as a guide for mission planning. The computation procedures shall include normal and, if required, alternate bomb method solutions. Ballistic graphs, bomb tables, and trigonometric tables shall be included. Ballistic graphs shall be in the format required by the applicable aircraft/bomb combination. Error analysis shall be included if applicable.

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3.2.4.1 Arrangement-Volume III (ballistics). This manual shall have the following arrangement:

Front Matter
Section I - Definition of Terms
Section II - Bombing Data
Section III - Bombing Data Forms

If, because of aircraft equipment, a particular section is not applicable, that section may be omitted and succeeding sections renumbered.

3.2.4.2 Front matter. Front matter shall be prepared as specified in 3.2.1 herein. In addition, the introduction shall include a statement concerning reproduction of the charts and graphs in the manual for in-flight use, based on mission essential requirements and complying with all classification marking/handling directives.

3.2.4.3 Section I - Definitions of terms. This section shall include definitions of abbreviations and terms related to the bombing system and tactics covered in the manual.

3.2.4.4 Section II - Bombing data. This section shall include ballistic tables and graphs used to compute information needed for bombing missions and instructions for their use. Ballistic graphs, when used, shall always be prepared on one side only (with blank backup pages). Pertinent instructions for use of each ballistics graph shall be placed on the same or facing page. (Note: Normally neither reproducible copy nor negatives will be prepared for blank pages.) Pages shall be numbered in accordance with MIL-M-38784.

3.2.4.5 Section III - Bombing data forms. This section shall contain instructions for preparation of bombing data forms for all applicable mission types. Sample filled-in forms shall also be included.

3.2.5 Checklists. If required, checklists shall be prepared in accordance with the requirements of MIL-C-38778 for type I, class 6 checklists. Checklist steps shall be based on the normal and emergency procedures in Volume II; steps shall consist of the unclassified demand-response line items from the amplified procedures; emergency data shall include tabular bomb malfunction analysis. Emergency pages shall be marked as specified in MIL-C-38778. (Note: The checklist will be used in conjunction with the aircraft flight crew checklists for nuclear missions.)

3.3 Practice bomb manual. This manual shall cover ballistics and computation data for practice bombing (Radar Bomb Sight-RBS) missions. The manual shall be titled "Practice Bomb Delivery - Ballistics." It shall be similar in scope to the ballistics volume of the nuclear delivery manual.

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3.3.1 Arrangement. The practice bomb manual shall have the following arrangement.

Front Matter
Section I - Bombing Data
Section II - Bombing Computation Forms

3.3.1.1 Front matter. Front matter shall be prepared as specified in 3.2.1 herein. The introduction shall include definition of terms used in the manual.

3.3.1.2 Section I - Bombing data. This section shall include ballistic tables, graphs, or charts required for the practice bombing mission. Tabbing and bomb delivery mode identification are not required in this manual.

3.3.1.3 Section II - Bombing data forms. This section shall contain instructions for preparation of bombing data forms and filled-in sample forms.

3.4 Maintenance of the manual. The manual (both procedural volumes and checklist) shall be maintained in accordance with the requirements of MIL-M-38784. Change and revision requirement shall be established as stated in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Basic Manual and Checklist

4.1.1 Preliminary review. Preliminary manuscripts shall be submitted for review. Comments and corrections resulting from this review shall be incorporated into the review copies.

4.1.2 Review Conference. A review conference shall be held at the contractor's facility, or if combined with a verification (reference TO 00-5-1), at an operational location. Review copies shall contain all data required by this specification and shall be in the form of manuscript (text double spaced) with art. The copy shall be reviewed for technical adequacy and accuracy. Before preparation of reproducible copy is begun, final manuscripts shall be approved for technical content and to ascertain that results of review/verification have been correctly incorporated into the copy.

4.1.3 Review of reproducible copy (prepublication review). Prior to preparation of negatives, reproducible copy shall be reviewed for technical accuracy and conformance to specifications.

4.1.4 Proof copies. When specified in the contract, proof copies shall be submitted as required on the DD Form 1423.

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4.2 Changes.

4.2.1 Preliminary review. Preliminary drafts of the manuscript for changes shall be submitted as required on the DD Form 1423 for review. Manuscript for changes shall be submitted for review. Manuscript for changes shall be clear and legible and shall include, as a minimum, a listing of existing art work to be used. Any changes in the art or requirements for new art shall be indicated. Comments and corrections resulting from this review shall be returned to the contractor for inclusion in the reproducible copy.

4.2.2 Review of reproducible copy and proof copies. Reproducible copy of changes shall be reviewed and proof copies furnished as specified for the basic manual.

4.3 Revisions. Revisions shall normally be prepared, reviewed and approved, and proof copies furnished as specified for the basic manual. Printed copies for the previous issue of the manual may be used for review/verification.

4.4 Negatives. Negatives shall meet the requirements of MIL-P-38790 and MIL-C-38778.

5. PREPARATION FOR DELIVERY

5.1 Negatives. Negatives shall be prepared for shipment in accordance with MIL-P-38790.

5.2 Security provisions. All shipments shall observe the applicable provisions of the current issue of the DoD Industrial Security Manual.

6. NOTES

6.1 Intended use. Technical Manuals and Checklists prepared to this specification are for use of strategic bomber aircrews in the level delivery of nuclear weapons.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Form in which the manual and checklist shall be furnished (repro copy, negatives).

6.3 TO numbering. The TO numbering information is presented here for information only since assignment of TO numbers is governed by AF regulations. The TOs are numbered in the aircraft -25 series, with the "dash 25" indicating nuclear bomb delivery. For example:

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Volume I - TO 1B-XX-25-1
Volume II - TO 1B-XX-25-2
Checklist - TO 1B-XX-25-2CL-1
Volume III - TO 1B-XX-25-3
Practice Bomb Manual - TO 1B-XX-25-10

Custodian
Air Force - 15

Preparing Activity
Air Force - 15

Review Activity
Air Force - 16, 79, 01, 10

Project No. TMSS-F323

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